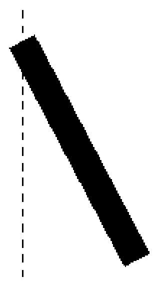
1 The Rod dangling from a Flywheel - 15 marks

We all are familiar with the simple pendulum - where one suspends a point mass from a massless rigid inextendible thread of a fixed length. The system behaves like a simple harmonic oscillator only under small angular displacement of the vertical direction.

If one has a massive rod suspended from one of its points, one can still have pendulum-like behavior but one will have to handle to include the rotational motion of the distributed mass - which leads to the appearance of the moment of inertia in the expression for the timeperiod for small oscillations.

So as a warmup let us go through this situation: A uniform rod of mass M and length L is suspended and pivoted from a point which is at a distance x away from its center of mass of the rod. The rod is allowed to swing around that point and let us ignore the friction at the pivot as well as the air resistance.

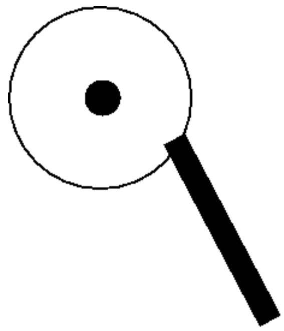


(a) If the rod is tilted at an angle θ , what is the torque acting on the rod which will try to orient the rod back to its vertical position. **1**

(b) If the moment of inertia for the rod about the suspended point is I_x , what will the timeperiod of oscillation for the rod in terms of the quantities provided above and the standard constants. **2**

(c) In the discussion above we did not demand that the rod be pivoted above the center of mass. So, one has to allow for both the possibilities where the rod is "hung" either below or above the center of mass of he rod. Considering both these possibilities,sketch the graph which shows how the timeperiod of (b) will vary as a function of the distance x. **3**

The game changes In the above we treated the point of suspension for the rod to be a fixed point. However, for some whimsical reason or another, let us decide to treat that point to lie on the the rim of a solid disk of radius R and mass m which itself can rotate about its center. Let the initial location of the point of suspension be horizontal and the rod itself be vertical. A generic configuration of the system is shown in the accompanying figure below.



(d) Find the coordinate of the tip of the suspended rod if the angular displacement of the rod from its vertical position is θ while the disk undergoes a rotation by an angle ϕ **2**

(e) What is the total energy of the system is terms of the variables given above. **3**

(f) Find the equations of motion for the system in terms of the appropriate variables. **4**

2 Collapsing Cloud-21 Marks

All stars have their beginning in cloud of dusts which collapses under the action of its own gravity. We will look at this collapse process in this problem. The word dust used for particles which are so slowly moving that they exert no process. Let us assume that the cloud is spherical with an initial radius r_0 and is of mass M. We assume that during the collapse (or contraction) the cloud remains spherical and does not have have any angular motion. We assume energy is conserved during this process which is given by the initial gravitational potential energy as the dust doesn't have any kinetic energy at the beginning.

(a) What is the expression for the initial energy for a dust particle of mass m under the conditions given? **1**

(b) What will be the expression for energy of the same particle, when the cloud contracts to a radius r? **2**

(c) Using the results from (a) and (b) find the integral that expresses the time required for the cloud to contract from the initial radius r_0 to the radius r_1. **3**

(d) Given the integral

$$\int_0^a \frac{dx}{\sqrt{\frac{1}{x} - \frac{1}{a}}} = (\frac{\pi}{2})a^{\frac{3}{2}}$$

Evaluate the time required for the cloud to collapse completely. **3**

(e) Compare your result with the time required for a satellite to revolve around the original cloud skimming its surface - i.e. the concentric orbit with a radius equal to r_0. **3**

However as the cloud collapses the gas heats up leading to an increase in pressure. In our discussiion above we have neglected this pressure buildup. This pressure will try to halt the dust from collapsing. As long as the initial energy of the cloud is greater than the internal energy of the gas - the cloud will keep on collapsing.

(f) What is an estimate for the gravitational energy of the cloud? **2**

(g) If the temperature of the cloud is T and the mass of individual particles making up the cloud is μ - find the kinetic energy of the cloud. **1**

(h) Using the condition for the collapse stated above find density below which the cloud will fail to condense. **4**

(i) If the mass of the sun is 2×10^{30} kg , find the density below which a cloud of molecular hydrogen of 20,000 Solar mass at a temperature of 20K will fail to condense. **2**

3 Is this Data-Science?- 9 Marks

(i) A student measures a certain length using a very precise method and ends up recording the result as 1.89999679 ± 0.00346 m. His supervisor asks him to round up his result and rewrite it in the report. What is the correct result that he must record? **2**

(ii) A certain theory predicts the lifetime of neutron to be 910 secs. However, experimental results show that the neutron lifetime is 950 ± 20. How many "sigmas" discrepancy is the theory with the experiment? **2**

(iii) The following table shows the attendance and the GPA of several students in a course. The teacher wants to propose the idea that higher attendance leads to higher GPA. How would you proceed to validate his hypothesis? Show your work in detail. No work shown will lead to no credits awarded. **5**

GPA	Days attended
4.00	180
2.50	150
4.00	170
3.90	180
3.75	177
3.80	180
2.90	140
3.10	169
3.25	168
3.40	152
3.30	150
3.90	170
1.35	109
4.00	180
1.00	108